their collaborator Wolfgang Pauli, are responsible for the Copenhagen interpretation of quantum mechanics, though who exactly believed what is a topic of ongoing historical and philosophical debate.

Heisenberg's approach in 1926, reflecting the boldness of a younger generation coming on the scene, was to put aside questions of what was really happening in a quantum system, and to focus exclusively on explaining what was observed by experimenters. Bohr had posited quantized electron orbits without explaining why some orbits were allowed and others were not. Heisenberg dispensed with orbits entirely. Forget about what the electron is doing; ask only what you can observe about it. In classical mechanics, an electron would be characterized by position and momentum. Heisenberg kept those words, but instead of thinking of them as quantities that exist whether we are looking at them or not, he thought of them as possible outcomes of measurements. For Heisenberg, the unpredictable jumps that had bothered Rutherford and others became a central part of the best way of talking about the quantum world.

Heisenberg was only twenty-four years old when he first formulated matrix mechanics. He was clearly a prodigy, but far from an established figure in the field, and wouldn't obtain a permanent academic position until a year later. In a letter to Max Born, another of his mentors, Heisenberg fretted that he "had written a crazy paper and did not dare to send it in for publication." But in a collaboration with Born and the even younger physicist Pascual Jordan, they were able to put matrix mechanics on a clear and mathematically sound footing.

It would have been natural for Heisenberg, Born, and Jordan to share the Nobel Prize for the development of matrix mechanics, and indeed Einstein nominated them for the award. But it was Heisenberg alone who was honored by the Nobel committee in 1932. It has been speculated that Jordan's inclusion would have been problematic, as he became known for aggressive right-wing political rhetoric, ultimately becoming a member of the Nazi Party and joining a *Sturmabteilung* (Storm trooper) unit. At the same time, however, he was considered unreliable by his fellow Nazis, due to his support for Einstein and other Jewish scientists. In the end, Jordan never won the prize. Born was also left off the prize for matrix mechanics, but that omission was made up for when he was awarded a separate Nobel in 1954 for his formulation of the probability rule. That was the last time a